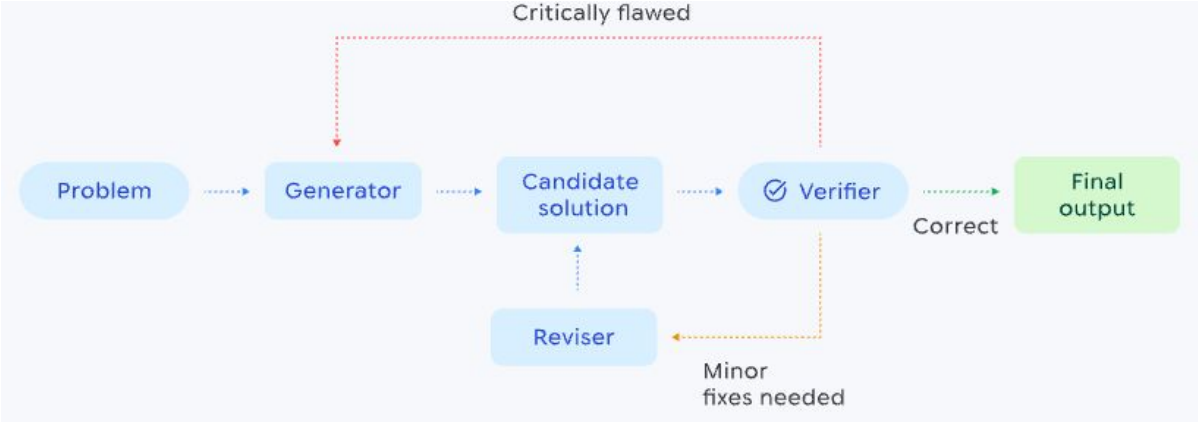
LLM usage in pure maths: what works and what doesn't

Fetch.ai Innovation Lab, 26 Mar 2026

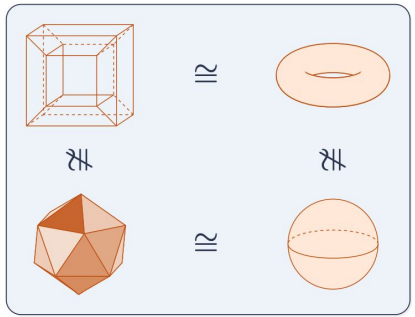
I will give an informal overview on how LLMs are currently used by pure maths researchers. While I will mention some high-flying niche applications, I will mostly focus on how an ordinary mathematician uses LLMs. Using examples, I will explain what already works well, and what some current problems are.

Some cutting edge LLM+pure maths research

- FunSearch/AlphaEvolve: counter-example generation [mostly] in combinatorics
- Aletheia



- LLM+Lean (AlphaProof, Aristotle, Axiom)
- Cambridge: “Discovering mathematical concepts through a multi-agent system”



Images: https://deepmind.google/blog/accelerating-mathematical-and-scientific-discovery-with-gemini-deep-think/
https://arxiv.org/pdf/2603.04528

Good: Literature search (best for materials available online)

- Anon. Math.: “Sure I use AI. It’s basically Google isn’t it?”
- “Please give a textbook reference for the elliptic estimate ||df||{C^{0,\alpha}(B_1)} \leq c(||\Delta f||{C^0(B_2)}+||f||_{C^0(B_2)}.”

	Answer	Reference
ChatGPT:	The cleanest direct reference I found is not a textbook but Viaclovsky’s MIT lecture notes. In Lecture 8, Proposition 1 proves directly from the Newtonian potential that bounded data give a $C^{1,\alpha}$ potential, and Theorem 1 plus Corollary 1 then yield the interior estimate $$\|u\|_{C^{1,\alpha}(\Omega')} \leq C\big(\|u\|_{C^0(\Omega)} + \|\Delta u\|_{L^\infty(\Omega)}\big), \qquad 0 \leq \alpha < 1,$$	**Corollary 1** *Suppose $\Delta u = f, f \in L^\infty(\Omega), \Omega' \subset\subset \Omega$, then for $0 \leq \alpha < 1$,* $$\|u\|_{C^{1,\alpha}(\Omega')} \leq C(\|u\|_{C^0(\Omega)} + \|f\|_{L^\infty(\Omega)}).$$ ✅
Fetch.ai:	**1. Gilbarg & Trudinger, "Elliptic Partial Differential Equations of Second Order" [1]** • **Section 4.6: Schauder Estimates** • Theorem 4.17 gives the boundary Schauder estimate for Laplace's equation	**Theorem 4.16.** *Let $u \in C^0(\bar{B}_2^+)$ satisfy Poisson's equation (4.33), where $f \in C^\alpha(\bar{B}_2^+)$, and suppose $u = 0$ on $B_2 \cap \{x_n = 0\}$. Then* $$(4.43) \qquad |u|'_{1,\alpha;B_1^+} \leqslant C(|u|_{0;B_2^+} + R|\mathbf{f}|'_{0,\alpha;B_2^+}), \qquad C = C(n,\alpha).$$ [Last theorem in this section] ❌
Gemini:	**Textbook References** • **Jürgen Jost, *Partial Differential Equations***: The direct integral splitting technique used to bound the fundamental solution is explicitly detailed here (often referenced around Theorem 13.1.1, which details the logarithmic modulus of continuity for the Newtonian potential of an L^∞ function).	$$\|u\|_{C^{1,\alpha}(\Omega)} \leq c_1 \sup\|f\| \quad \text{for } \alpha \in (0,1).$$

Good: Questions where I'm no expert

- "The Polymath Companion" (example 1, example 2)
- Example: "Consider the boundary value problem [...]. Please prove the interior elliptic estimate for it."
  (Typical undergraduate maths homework question;
  3-4 hours for student taking the lecture;
  Near Impossible for me, not having taken the lecture)
- LLM produces full proof with some

(i) small logical gaps

You are right. My previous use of (10.154) on the **raw** operator was wrong.

(ii) needless complications

Yes. For the fixed-L coercive estimate, Taylor is the cleaner route, and the Grubb machinery is not needed.

(iii) notational errors

You are right to press this point. My earlier "square" remark was sloppy.

Bad: Questions that need attention to detail

- Example: “Let F:U->R^{n \times n}. Assume ||F_{ij}-δ_{ij}||_{C^{1,\alpha}(U)} < ε. What’s the sharpest constant in the estimate ||F^{ij}-δ_{ij}||_{C^{1,\alpha}(U)} < M?”*

ChatGPT:

$$M(\epsilon) = \frac{\epsilon}{1 - n\epsilon} + \frac{2\epsilon}{(1 - n\epsilon)^2} + \frac{2^{2-\alpha} n\epsilon^2}{(1 - n\epsilon)^3}$$

❓

Fetch.ai:

$$M = \frac{\epsilon}{1-\sqrt{n\epsilon}}$$

❌

Gemini:

The space $C^{1,\alpha}(U)$ is a Banach algebra. This means there exists a constant $C \geq 1$ (which depends on the domain U and the specific definition of your norm) such that $\|fg\|_{C^{1,\alpha}} \leq C\|f\|_{C^{1,\alpha}}\|g\|_{C^{1,\alpha}}$.

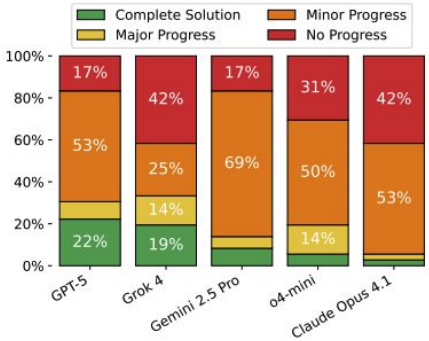
$$M = \sum_{k=1}^{\infty} n^{k-1} C^{k-1} \epsilon^k = \frac{\epsilon}{1 - nC\epsilon}$$

❌

*Full prompt: Let F:U->R^{n \times n} be a matrix-value function defined on an open set U \subset R^n. Assume that for \epsilon>0, small enough, we have the bound
||F_{ij}-\delta_{ij}||_{C^{1,\alpha}(U)} < \epsilon.
Then the inverse matrix F^{-1} exists on U and satisfies
||F^{ij}-\delta_{ij}||_{C^{1,\alpha}(U)} < M.
Find the sharpest bound for M in terms of \epsilon, and specify what "small enough" means.

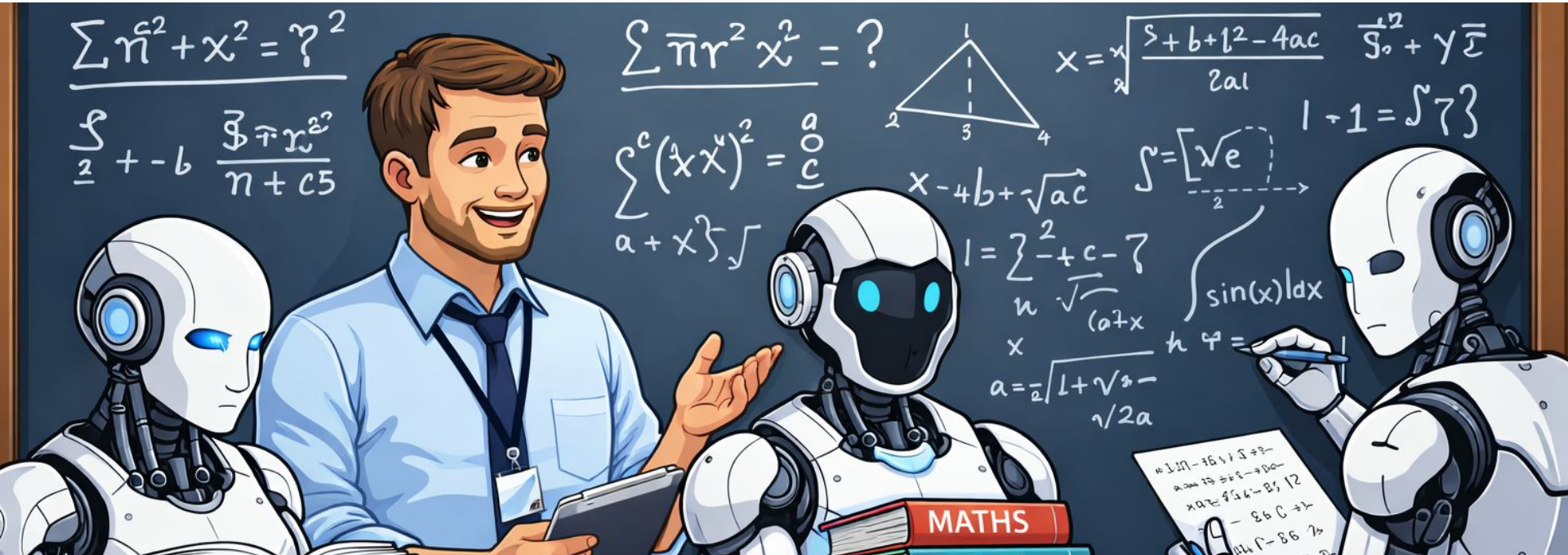
Bad: Questions where I am the expert

- IMProofBench: secret research level maths questions



- Anon. Math.: “If LLMs said ‘I don’t know’ rather than producing wrong maths, that would be a game changer.”
- Takes much time to find error in reasoning → usually lose time
- Exception: idea generation and finding connections to other areas of maths

Image: https://arxiv.org/pdf/2509.26076

Summary

- Mathematicians use LLMs
- Most useful as polymath and literature searchers
- Not good at details or very specialised questions
- Main blocker: wrong mathematics is worse than “I don’t know”